



CrysDiS

Crystal Diffraction Simulator

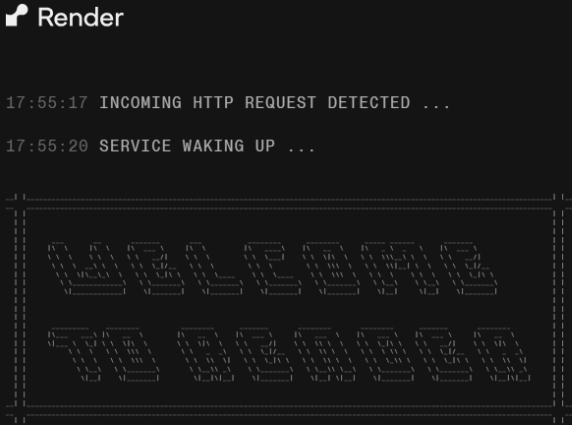
(created using ChatGPT Codex)



Start the program

Option 1: use the website version

- Website: <https://crysdiss.onrender.com/>
- Since this is a free URL, it may take 2~3 minutes to warm up
- It also does not have high computational performance
- If you think this software is useful, running it locally is strongly recommended



Render

17:55:17 INCOMING HTTP REQUEST DETECTED ...

17:55:20 SERVICE WAKING UP ...

17:55:24 ALLOCATING COMPUTE RESOURCES ...

17:55:27 PREPARING INSTANCE FOR INITIALIZATION ...

17:55:31 STARTING THE INSTANCE ...

17:55:37 ENVIRONMENT VARIABLES INJECTED ...

17:55:39 FINALIZING STARTUP ...

17:55:41 OPTIMIZING DEPLOYMENT ...

17:55:43 STEADY HANDS. CLEAN LOGS. YOUR APP IS ALMOST LIVE ...

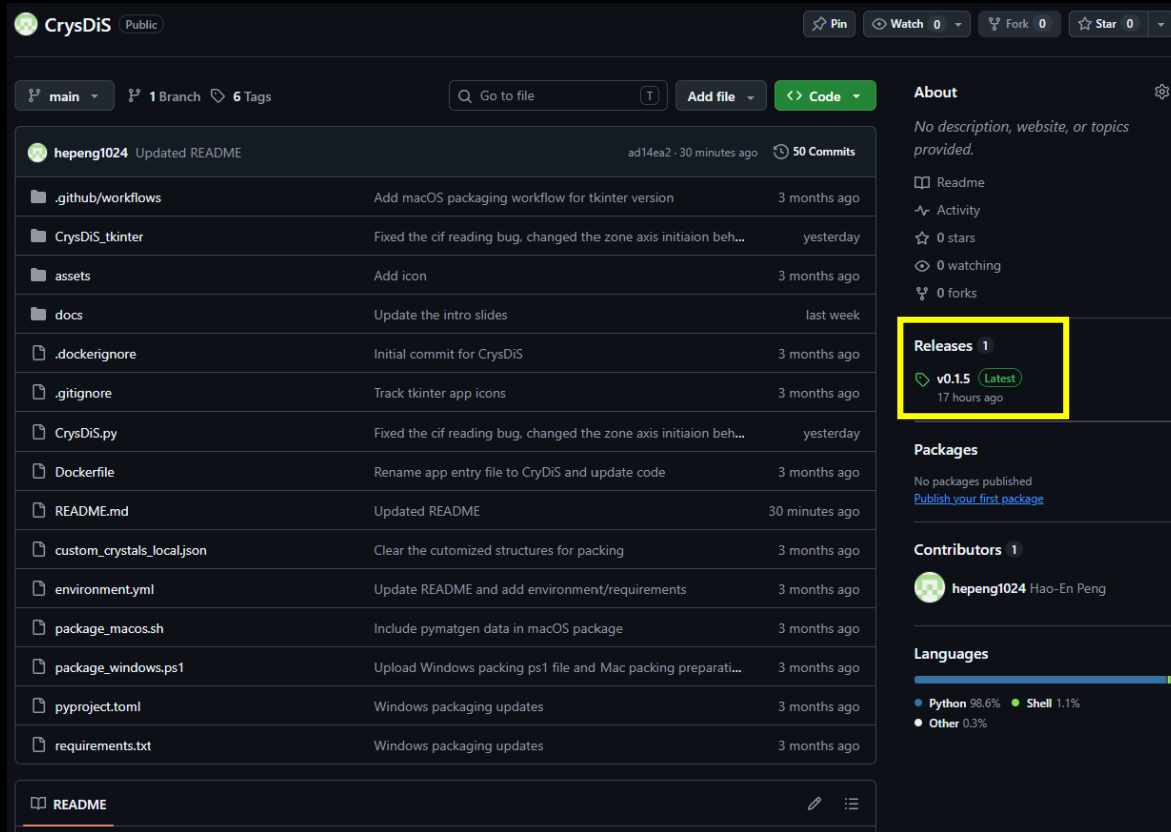
START BUILDING ON RENDER TODAY ->

APPLICATION LOADING

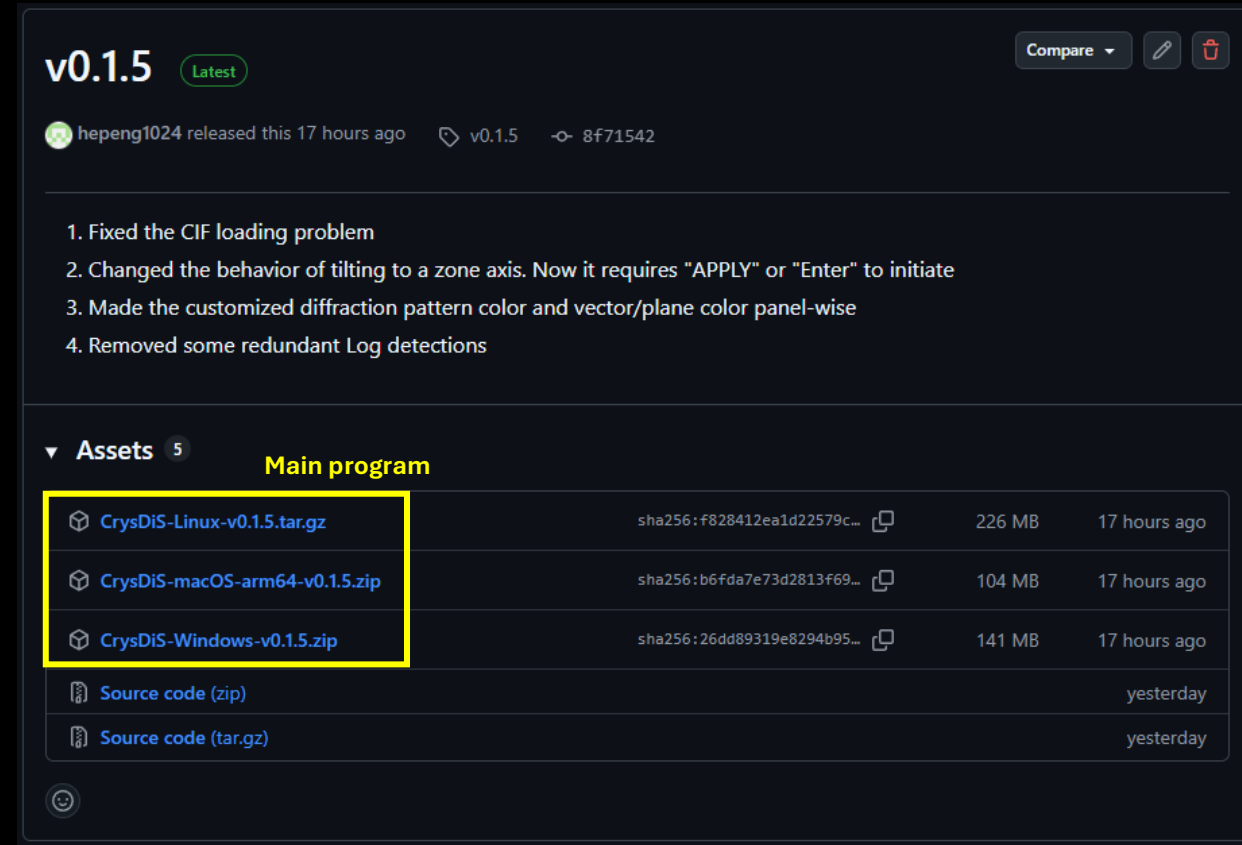
Start the program

Option 2: download the program and run it locally

- Go to <https://github.com/hepeng1024/CrysDiS> → Click “Releases” → Download the version that matches your OS



The screenshot shows the GitHub repository for CrysDiS. The 'Releases' section is highlighted with a yellow box, indicating the latest version is v0.1.5, released 17 hours ago. The repository includes a README, a list of files (e.g., .github/workflows, CrysDiS_tkinter, assets, docs, .dockerignore, .gitignore, CrysDiS.py, Dockerfile, README.md, custom_crystals_localjson, environment.yml, package_macos.sh, package_windows.ps1, pyproject.toml, requirements.txt), and a list of contributors.



The screenshot shows the release page for v0.1.5 of CrysDiS. The release notes are listed below the version number, and the assets section is highlighted with a yellow box, showing three download links: CrysDiS-Linux-v0.1.5.tar.gz, CrysDiS-macOS-arm64-v0.1.5.zip, and CrysDiS-Windows-v0.1.5.zip. The assets section also includes source code (zip) and source code (tar.gz) links.

Release Notes:

1. Fixed the CIF loading problem
2. Changed the behavior of tilting to a zone axis. Now it requires "APPLY" or "Enter" to initiate
3. Made the customized diffraction pattern color and vector/plane color panel-wise
4. Removed some redundant Log detections

Assets:

Asset Name	SHA256	Size	Time
CrysDiS-Linux-v0.1.5.tar.gz	sha256:f828412ea1d22579c...	226 MB	17 hours ago
CrysDiS-macOS-arm64-v0.1.5.zip	sha256:b6fda7e73d2813f69...	104 MB	17 hours ago
CrysDiS-Windows-v0.1.5.zip	sha256:26dd89319e8294b95...	141 MB	17 hours ago

- Please refer to the README file if you need more information

Start the program

Option 3: Run the source code from GitHub

- First install Git and Anaconda/Miniconda. Then clone the repository:

```
git clone https://github.com/hepeng1024/CrysDiS.git
```

```
cd CrysDiS
```

- Create the conda environment:

```
conda env create -f environment.yml
```

```
conda activate crysdis
```

- Start CrysDiS:

```
python CrysDiS.py
```

- Then open this address in a browser:

```
http://127.0.0.1:8080
```

- If port 8080 is already being used, choose another port:

```
NICEGUI_PORT=8094 python CrysDiS.py
```

- Please refer to the README file at <https://github.com/hepeng1024/CrysDiS> for more details

Starting interface

CrysDiS

[?](#) INTRO

☐ Current zone

☐ Spot indices

☒ Crystal annotations

☒ Scale bar

ADVANCED

CRYSTAL LIST

LOAD CIF

NEW CRYSTAL

ADD COMBO PANEL

ADD PANEL

1

Crystal
FCC

Zone axis

Plane

Vector

Rotation

APPLY

SYNC

0.2 nm

5 nm⁻¹

Features of the top toolbar

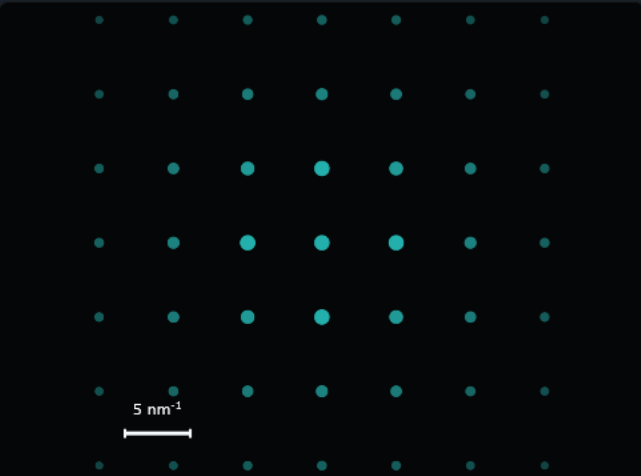
“ADD PANEL” allows you to have multiple structures at the same time

CrysDiS ? INTRO ☐ Current zone ☐ Spot indices ☒ Crystal annotations ☒ Scale bar ☒ ADVANCED ☒ CRYSTAL LIST ☒ LOAD CIF ☒ NEW CRYSTAL ☒ ADD COMBO PANEL ☒ **ADD PANEL**

1 Crystal: FCC Zone axis: Plane: Vector: Rotation: APPLY SYNC     



0.2 nm

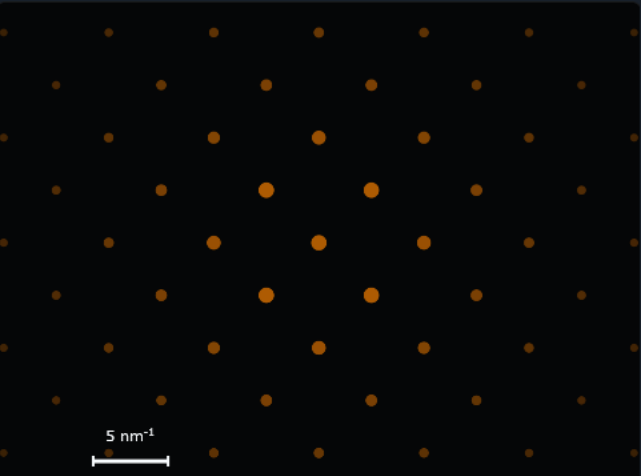


5 nm⁻¹

2 Crystal: BCC Zone axis: Plane: Vector: Rotation: APPLY SYNC     



0.2 nm

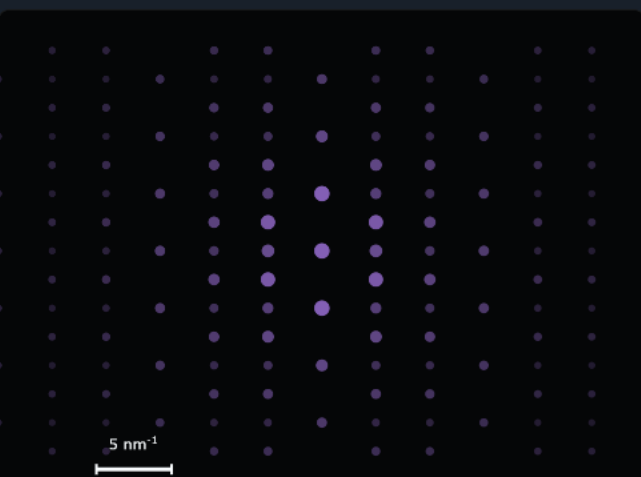


5 nm⁻¹

3 Crystal: HCP Zone axis: Plane: Vector: Rotation: APPLY SYNC     



0.5 nm



5 nm⁻¹

Features of the top toolbar

“ADD COMBO PANEL” allows you to overlap diffraction patterns from different panels and compare them

CrysDiS [?](#) [INTRO](#) ☐ Current zone ☐ Spot indices ☒ Crystal annotations ☒ Scale bar [ADVANCED](#) [CRYSTAL LIST](#) [LOAD CIF](#) [NEW CRYSTAL](#) **ADD COMBO PANEL** [ADD PANEL](#)

1 Crystal: FCC Zone axis: Plane: Vector: Rotation: [APPLY](#) [SYNC](#)

2 Crystal: BCC Zone axis: Plane: Vector: Rotation: [APPLY](#) [SYNC](#)

3 Crystal: HCP Zone axis: Plane: Vector: Rotation: [APPLY](#) [SYNC](#)

C1 Current crystals: 1. FCC [ADD](#) [REMOVE](#) ☐ Bind crystal motion

Panel	Crystal	Zone	
1	FCC	[100]	↑ ↓ ×
2	BCC	[100]	↑ ↓ ×
3	HCP	[2-1-10]	↑ ↓ ×

Features of the top toolbar

“NEW CRYSTAL” lets you build your own crystal

CrysDiS [INTRO](#) ☐ Current zone ☐ Spot indices ☒ Crystal annotations ☒ Scale bar [ADVANCED](#) [CRYSTAL LIST](#) [LOAD CIF](#) **NEW CRYSTAL** [ADD COMBO PANEL](#) [ADD PANEL](#)

Customized crystal builder

Name: Customized crystal

Lattice: cubic

Symmetry: 1: P1

Save target: User app-data library

a: 0.35 nm

b: 0.35 nm

c: 0.35 nm

alpha: 90 deg

beta: 90 deg

gamma: 90 deg

[APPLY LATTICE CONSTRAINTS](#) [+ ADD ATOM SITE](#) [PREVIEW SYMMETRY](#) [APPLY SAME COLOR TO SAME ELEMENT](#) [SET ELEMENT COLOR TO DEFAULT](#)

→ Apply the symmetry to the current structure

Atom sites

Atom: Ni

x: 0

y: 0

z: 0

Occ.: 1

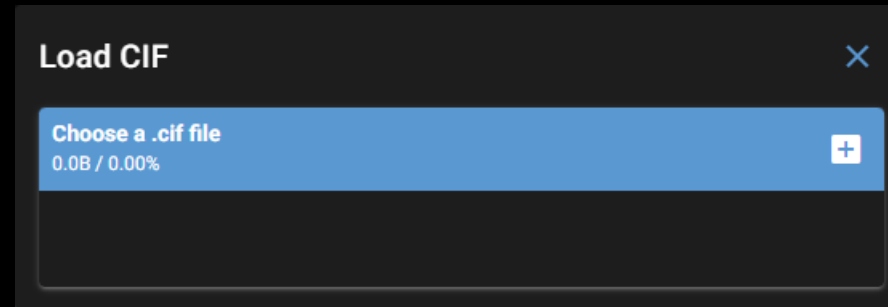
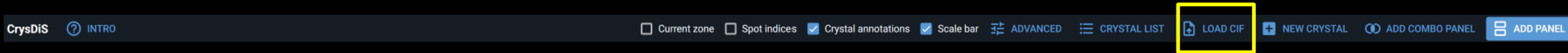
Color: #ed02e9

Site: Ni

[SAVE AS A NEW STRUCTURE](#) [EXPORT AS CIF](#)

Features of the top toolbar

You can load your own .cif files via “LOAD CIF”



Three built-in structures that cannot be edited

Crystal list

Structure	Download	Edit	Delete
FCC Built-in · cubic, Fm-3m, 1 site(s)			
BCC Built-in · cubic, Im-3m, 1 site(s)			
HCP Built-in · hexagonal, P1, 2 site(s)			
D024 (Ni,Mn,Fe)3Ti Custom · hexagonal, P6 ₃ /mmc, 8 site(s)			
D024 (Ni,Mn,Fe)3Ti_unequal_site_occ Custom · hexagonal, P6 ₃ /mmc, 8 site(s)			
D024 (Ni0.8Mn0.1Fe0.1)3Ti Custom · hexagonal, P6 ₃ /mmc, 8 site(s)			
D024 Ni3(Fe,Mn,Ti) Custom · hexagonal, P6 ₃ /mmc, 8 site(s)			
D024 Ni3(Fe,Mn,Ti)_unequal_site_occ Custom · hexagonal, P6 ₃ /mmc, 8 site(s)			
D024 Ni3(Fe0.1Mn0.1Ti0.8) Custom · hexagonal, P6 ₃ /mmc, 8 site(s)			
D024 Ni3Ti Custom · hexagonal, P6 ₃ /mmc, 4 site(s)			
D024 Ni3Ti (disordered) Custom · hexagonal, P6 ₃ /mmc, 8 site(s)			
D024 Ni3Ti (matproj_unsymmetrized) Custom · hexagonal, P1, 16 site(s)			
D024 Ni3Ti (matproj_unsymmetrized_distorted)			

Features of the top toolbar

Advanced settings in “ADVANCED”

CrysDiS INTRO

☐ Current zone ☐ Spot indices ☒ Crystal annotations ☒ Scale bar **ADVANCED** CRYSTAL LIST LOAD CIF NEW CRYSTAL ADD COMBO PANEL ADD PANEL

Advanced settings

Simulation method Pymatgen TEMCalculator	Camera length 200 mm	Voltage 500 kV
Thickness 10 nm	Max hkl 7	Spot intensity threshold 0.001
Compression factor 100 Contrast of diffraction spots intensity	Spot size scaling 0.3 How large bright spots are	☒ Always snap back view Check to rotate the crystal back when moving to a new zone
☒ Use four-index basis for hexagonal systems	☐ Auto sync crystal/diffraction	PALETTE FOR VECTORS/PLANES

Log

Ready

Added a comparison panel

Added a comparison panel

Added a combo panel

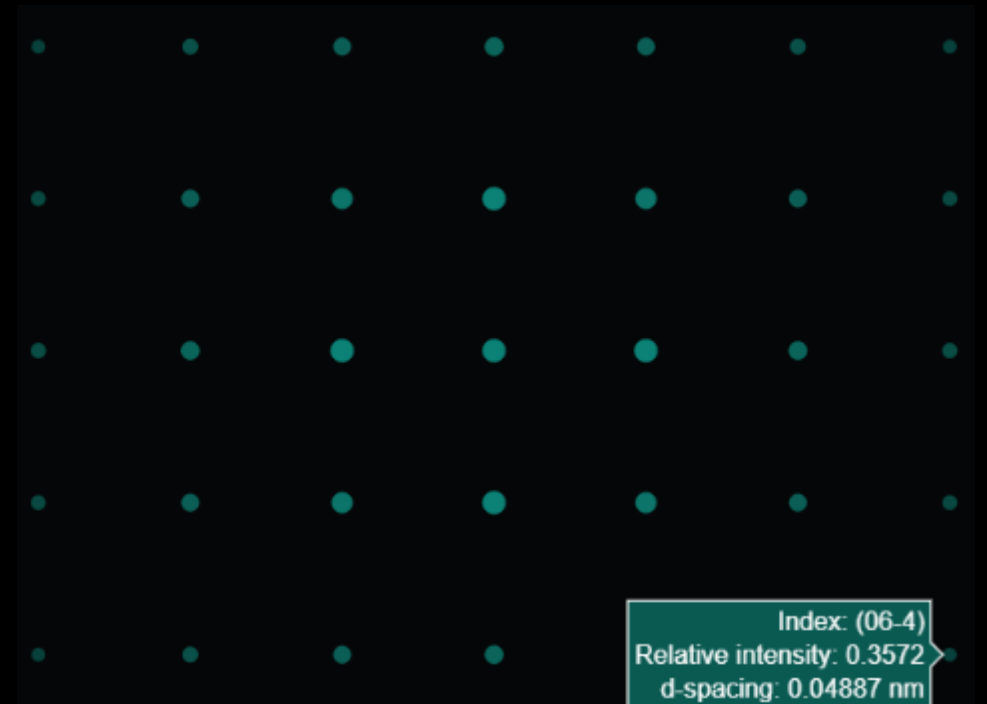
More about advanced settings

- Compression factor

Compression factor = 10



Compression factor = 150



Dim spots are more visible

More about advanced settings

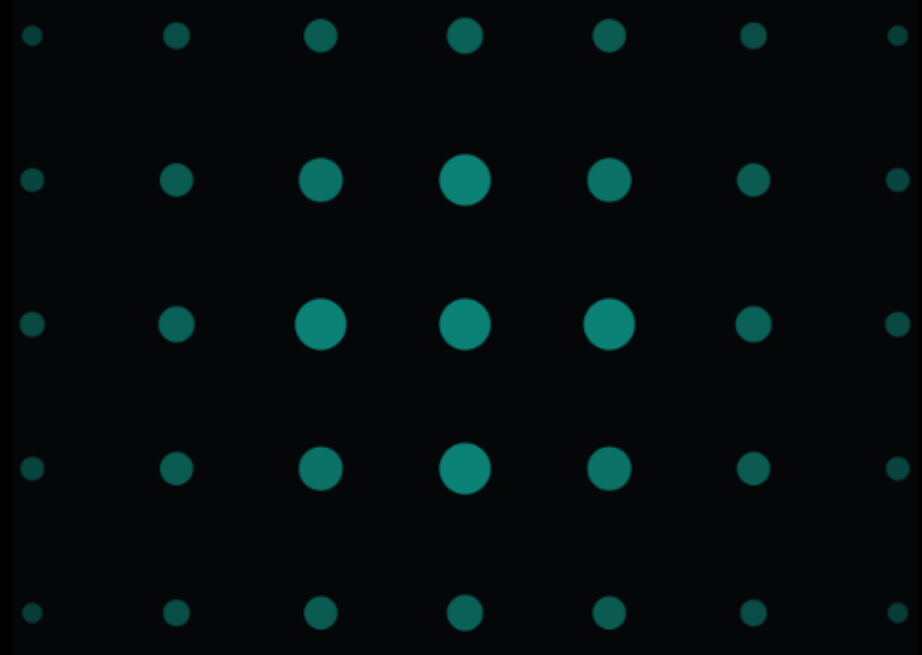
- Spot size scaling effect

Scaling factor = 0



All spots are the same size

Scaling factor = 1



Spot size scales significantly with intensity

More about advanced settings

- Always snap back view

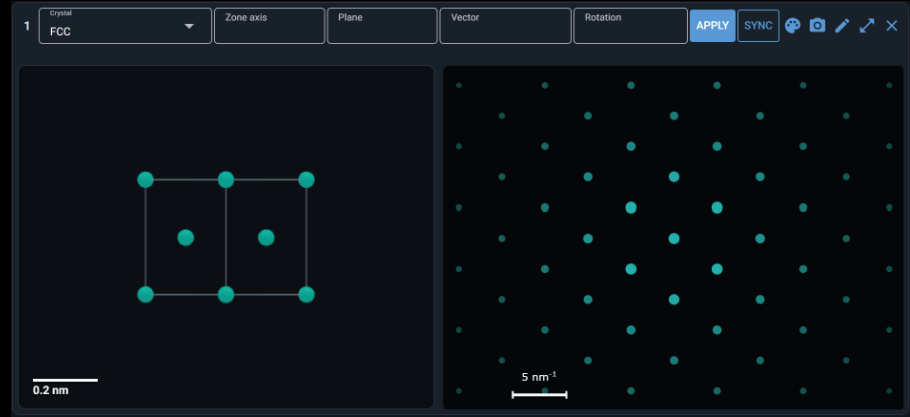


In-plane rotation by 45 deg



Go to the 110 zone

“Always snap back view” is checked



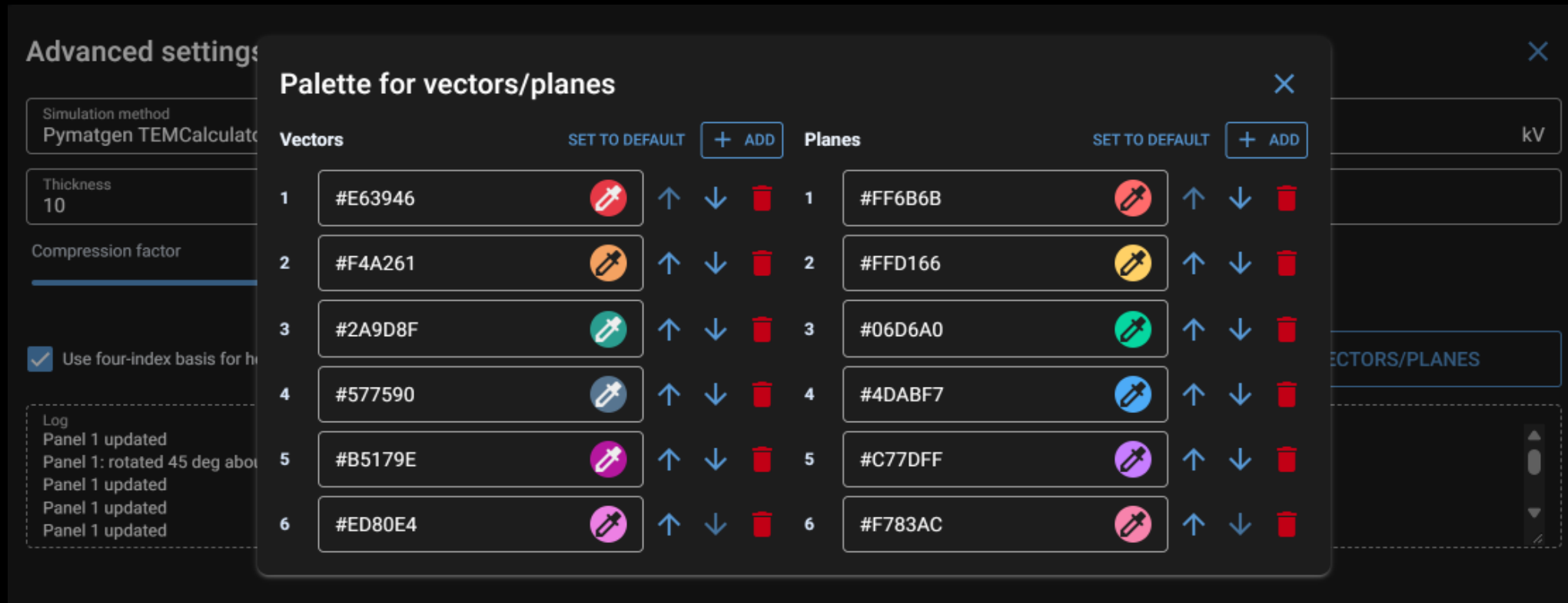
“Always snap back view” is unchecked



The in-plane 45 deg rotation history is retained

More about advanced settings

- You can also set the global colors of crystal vectors/planes
- Each panel will follow these color orders by default



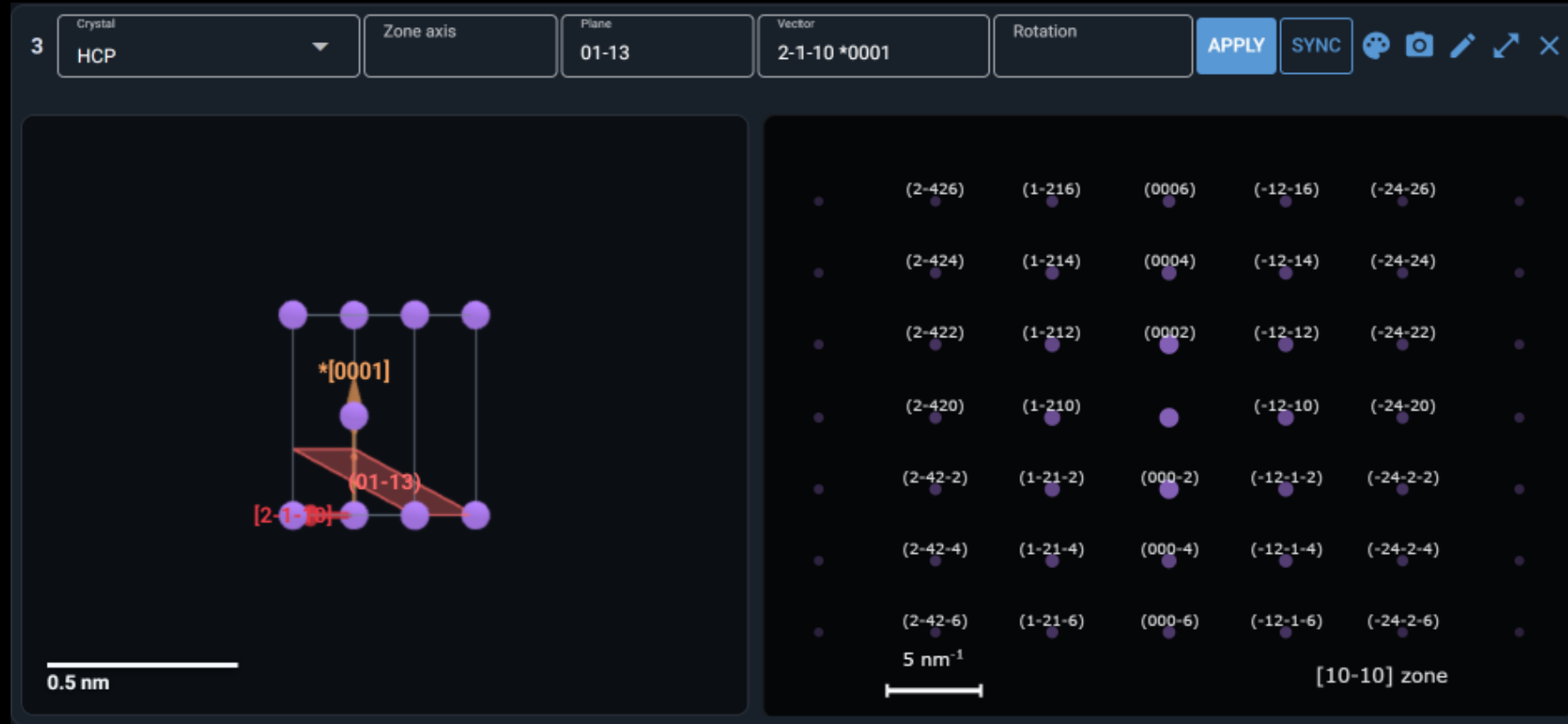
Features of the top toolbar

Some annotation toggles



Features of the panel toolbar

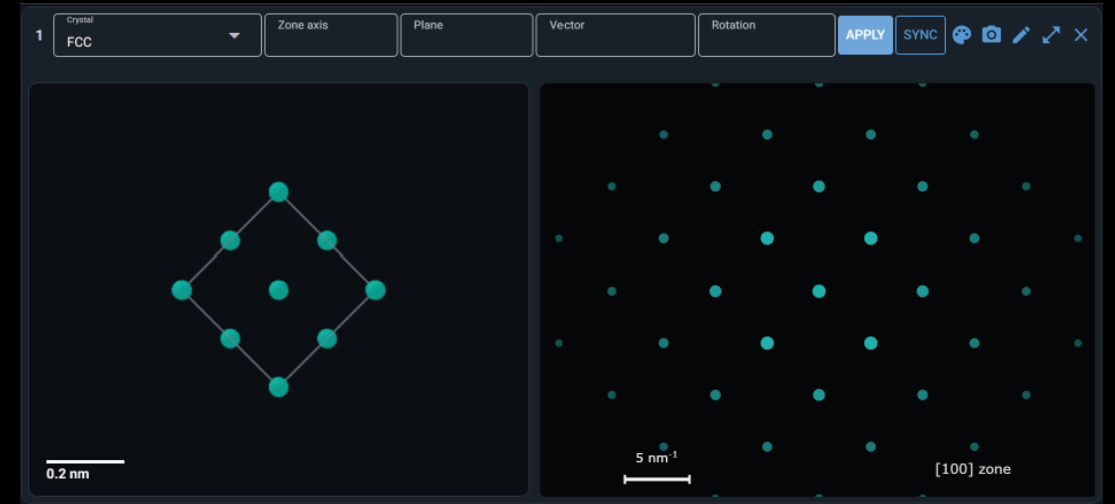
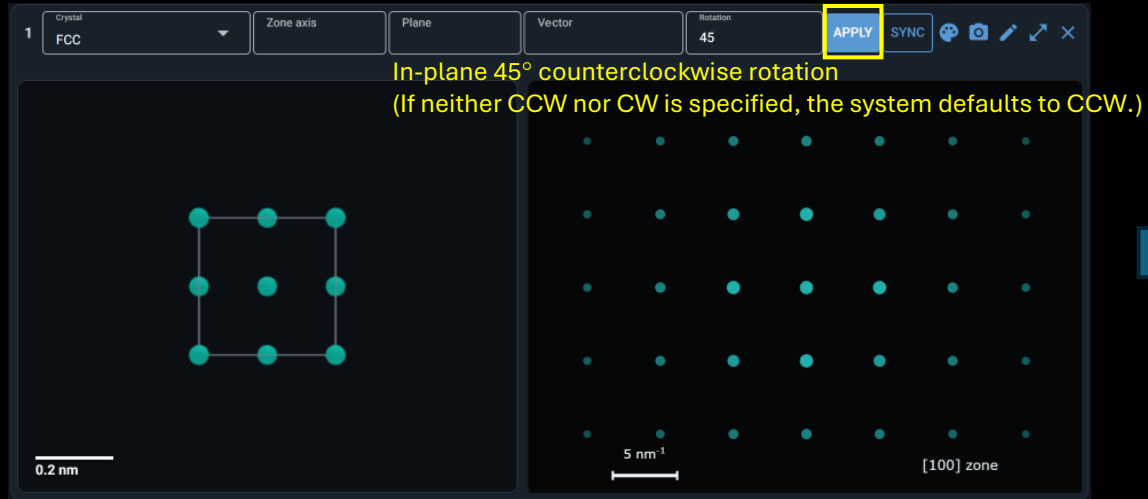
- Choose any crystals in the list or upload/build a new one
- You can go to any zone axis, create vectors/planes
- Reciprocal lattice vectors can be created using *hkl (ex. *110, *0-10)
- For hexagonal systems, 4-index basis inputs are allowed (ex. 2-1-10, *0001)



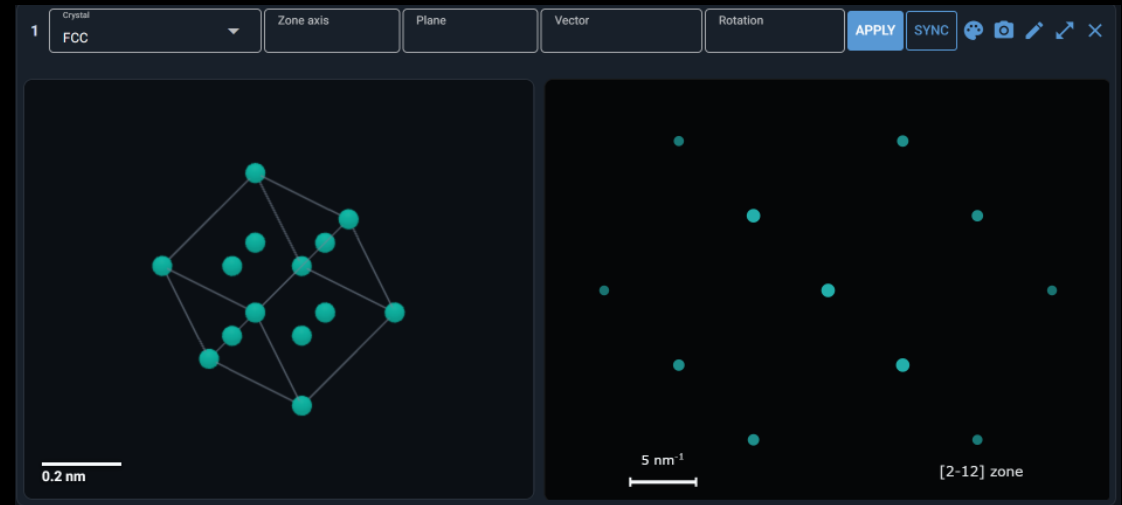
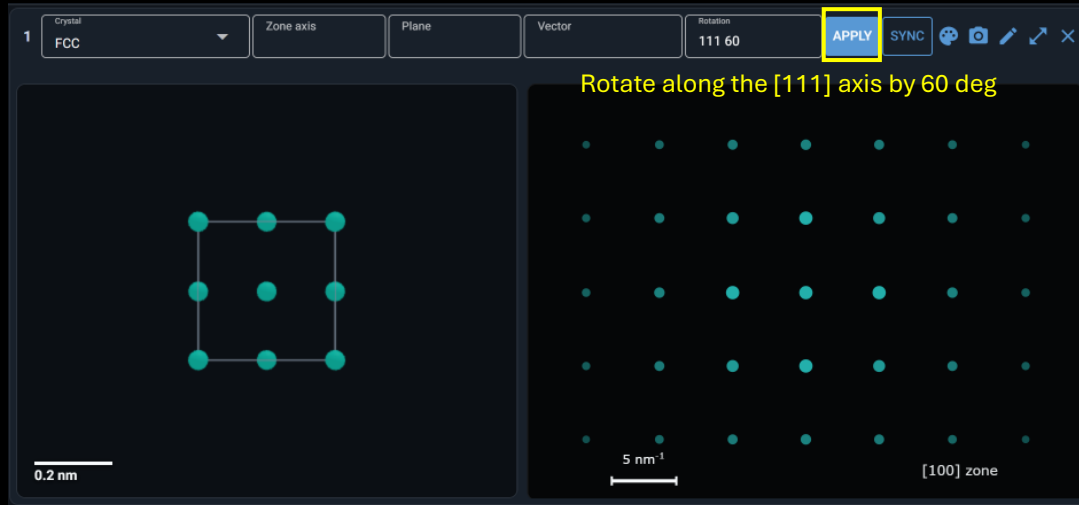
Features of the panel toolbar

- You can also rotate the crystal along a certain axis (right-hand rule is applied)

In-plane rotation: use “deg”, “deg cw”, or “deg ccw”

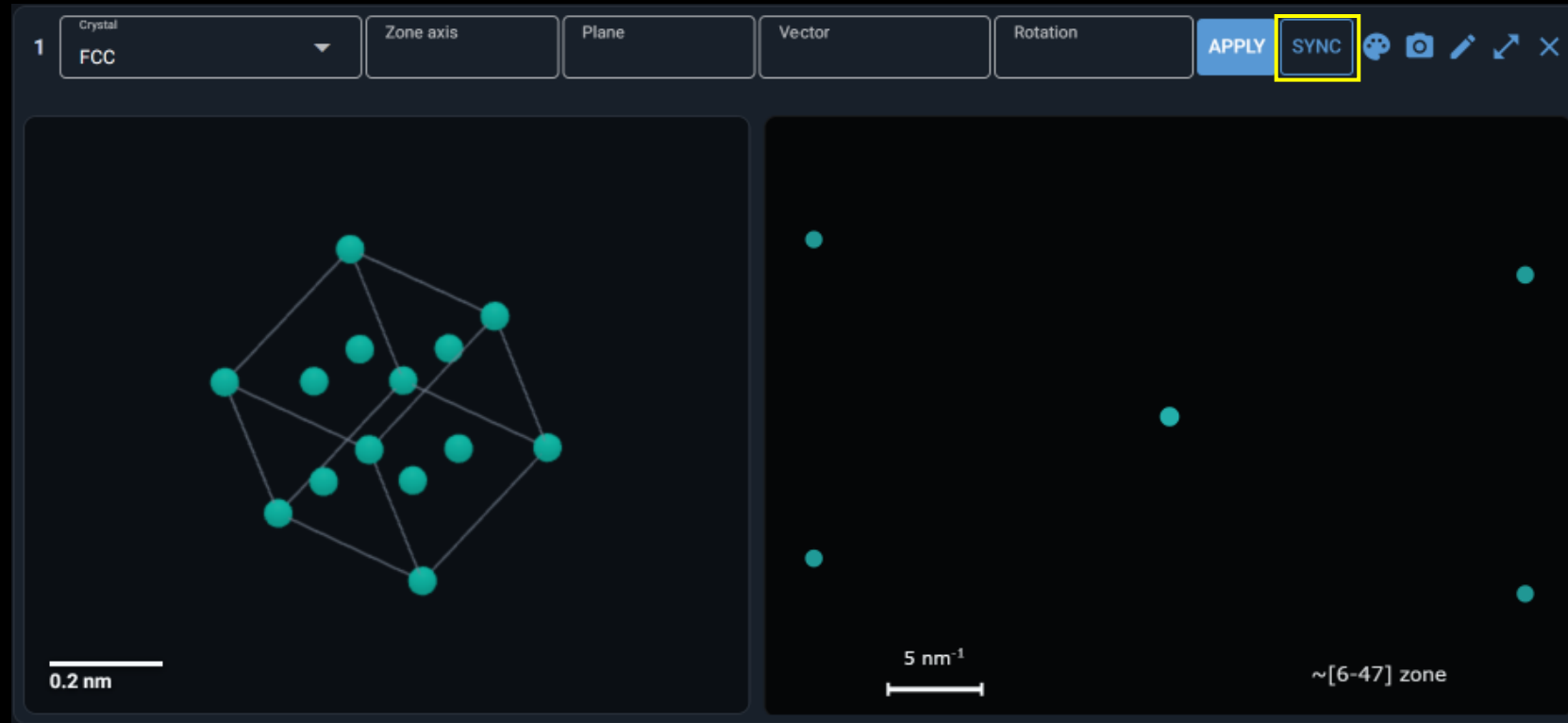


Rotation along a certain axis: use “vector deg”



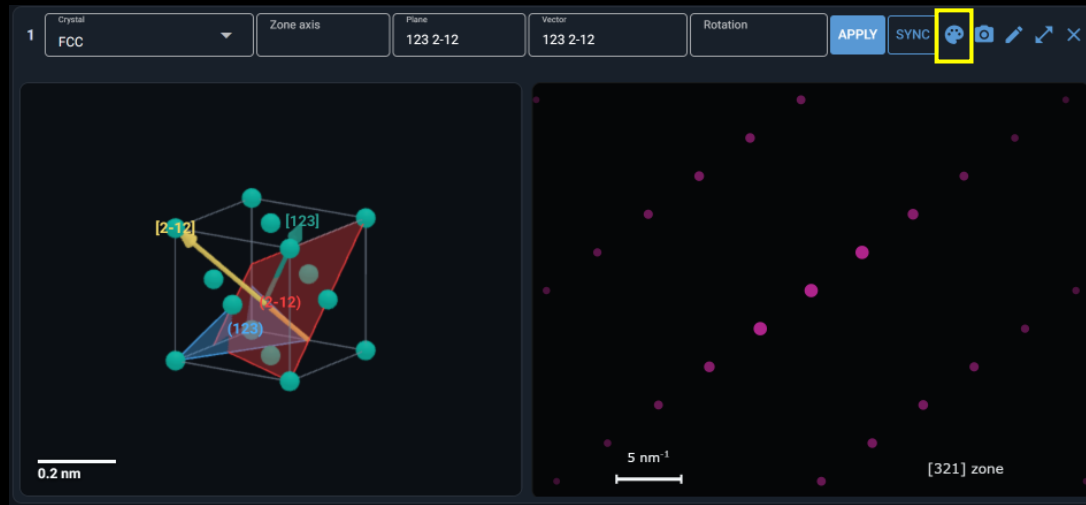
Features of the panel toolbar

- Synchronize the crystal and diffraction pattern using “SYNC”
- It will detect the closest zone axis
- You can also turn on the “Auto sync crystal/diffraction” in the advanced settings



Features of the panel toolbar

- You can customize each panel's diffraction pattern color and plane/vector colors
- This will overwrite the global default colors



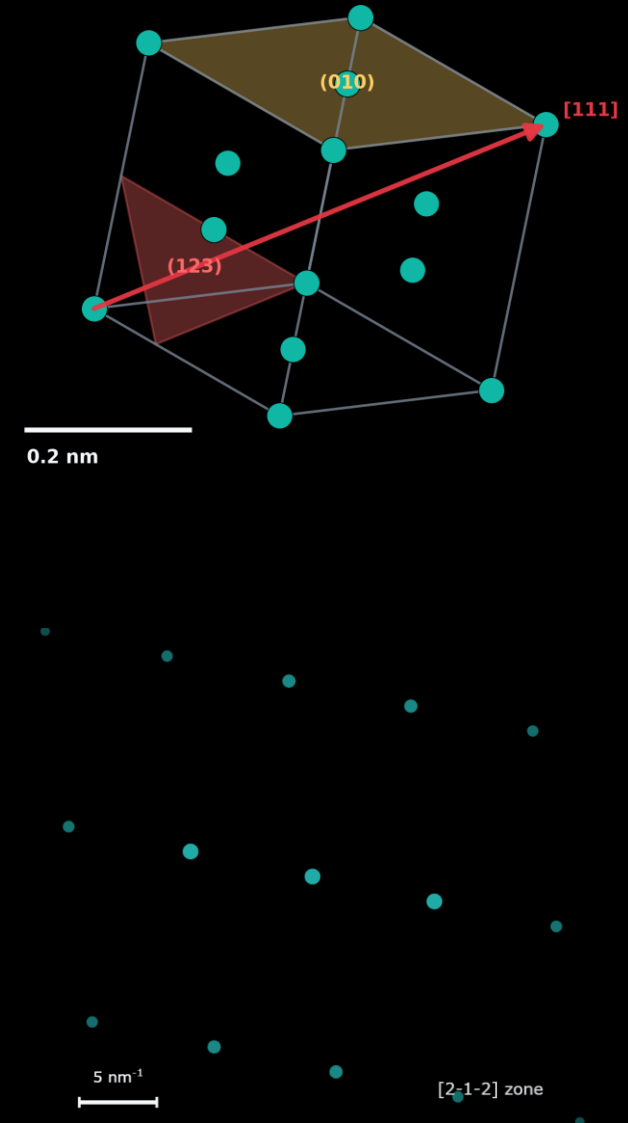
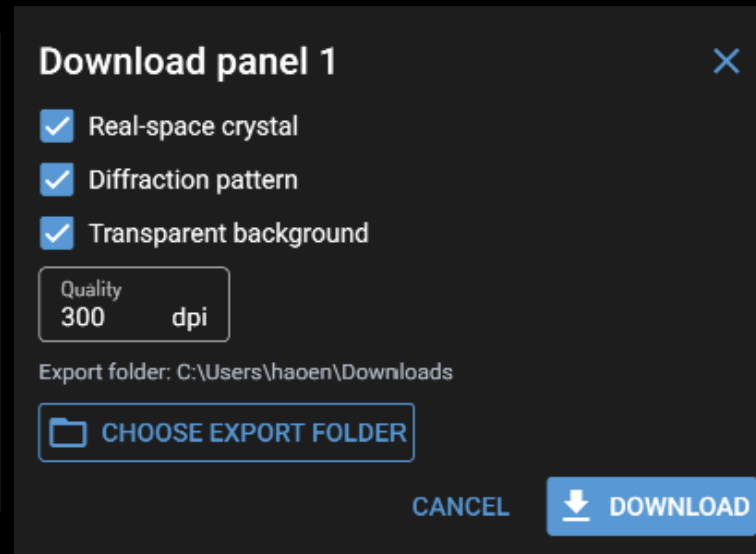
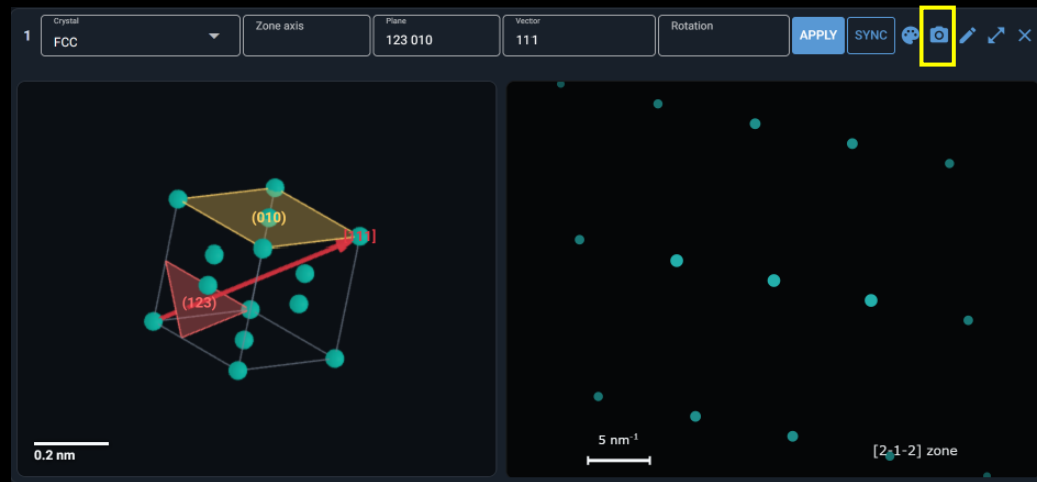
Panel 1 colors - FCC

Diffraction pattern #F731C3

Vectors				USE GLOBAL DEFAULT	+ ADD	Planes				USE GLOBAL DEFAULT	+ ADD
1	#2A9D8F		↑ ↓		1	#4DABF7		↑ ↓			
2	#EDD568		↑ ↓		2	#F74040		↑ ↓			
3	#E63946		↑ ↓		3	#C77DFF		↑ ↓			
4	#577590		↑ ↓		4	#FFD166		↑ ↓			
5	#B5179E		↑ ↓		5	#06D6A0		↑ ↓			
6	#ED80E4		↑ ↓		6	#F783AC		↑ ↓			

Features of the panel toolbar

- Download the crystal/diffraction pattern at the current viewing angle



Features of the panel toolbar



- Edit the current structure

Customized crystal builder

Name

L10 MnNi

Lattice

tetragonal

Symmetry

123: P4/mmm

Save target

User app-data library

a

0.374

nm

b

0.374

nm

c

0.352

nm

alpha

90

deg

beta

90

deg

gamma

90

deg

APPLY LATTICE CONSTRAINTS

+

ADD ATOM SITE

PREVIEW SYMMETRY

APPLY SAME COLOR TO SAME ELEMENT

SET ELEMENT COLOR TO DEFAULT

Atom sites

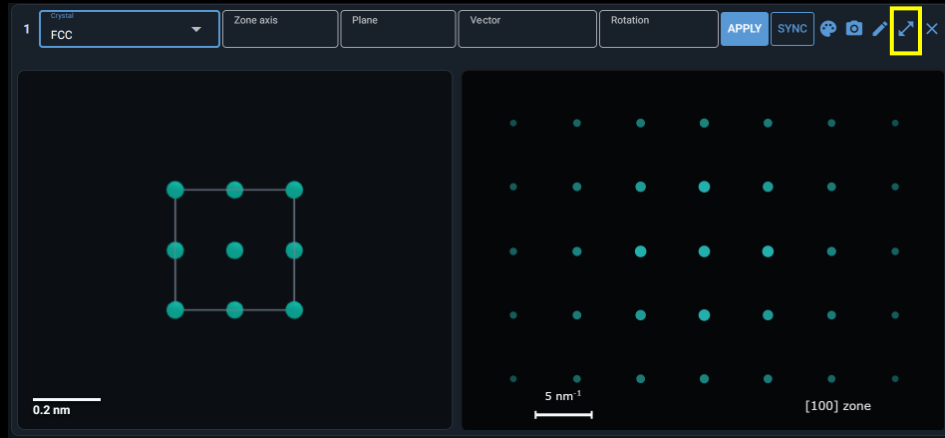
Atom Mn	x 0	y 0	z 0	Occ. 1	Color #dede00	Site Mn1
Atom Mn	x 0.5	y 0.5	z 0	Occ. 1	Color #dede00	Site Mn2
Atom Ni	x 0.5	y 0	z 0.5	Occ. 1	Color #ed02e9	Site Ni1

SAVE EDITED STRUCTURE

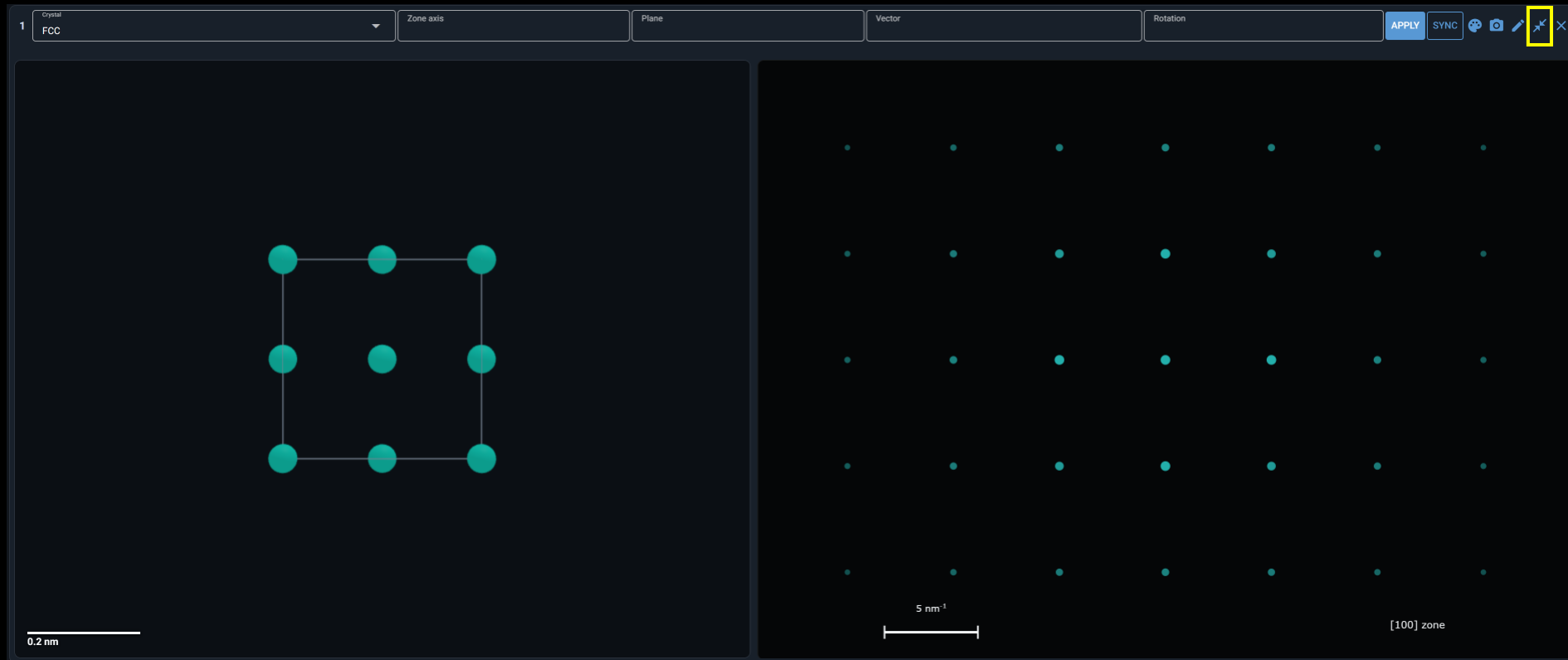
SAVE AS A NEW STRUCTURE

EXPORT AS CIF

Features of the panel toolbar



- Magnify the panel



- Shrink the panel back

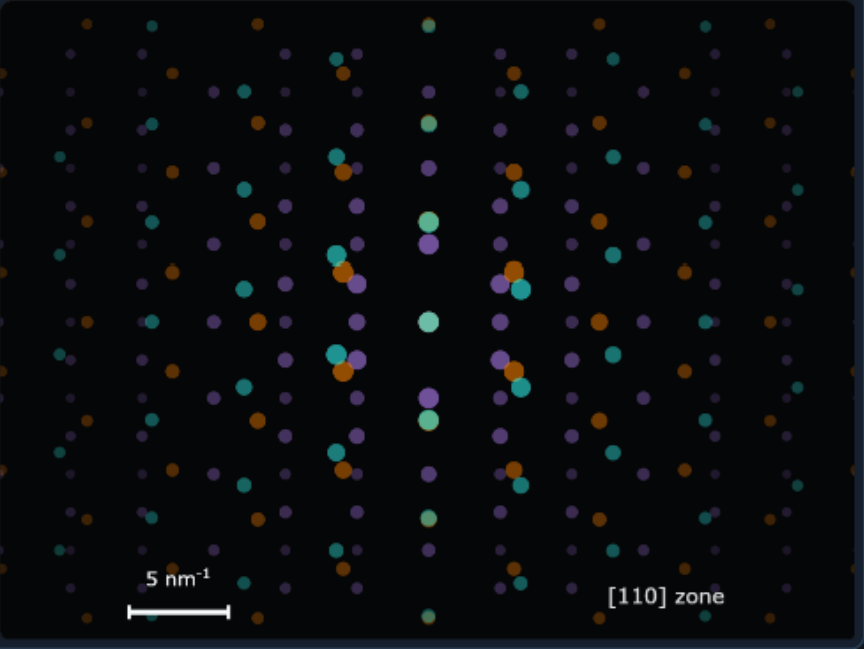
Features of the combo panel

Select crystals present on the page

Add or remove it

C1 Current crystals 1. FCC ADD REMOVE ☐ Bind crystal motion   

Panel	Crystal	Zone	
1	FCC	[110]	  
2	BCC	[111]	  
3	HCP	[2-1-10]	  



Ex. FCC diffraction pattern will be shown on top

The current zone shows the zone axis of the crystal on the top-most layer

The order of the crystal in the list determines the order of the diffraction patterns

Features of the combo panel

Check “Bind crystal motion” if you want the crystals to synchronize their motions

C1

Current crystals
1. FCC

ADDREMOVE☒ Bind crystal motion

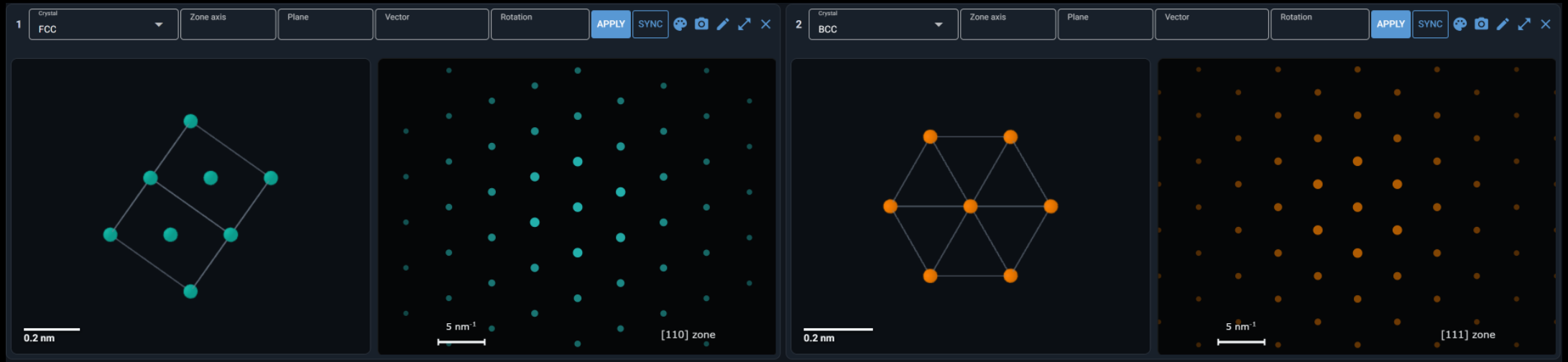
Panel	Crystal	Zone	
1	FCC	[110]	↑ ↓ ×
2	BCC	[111]	↑ ↓ ×

5 nm⁻¹

[110] zone

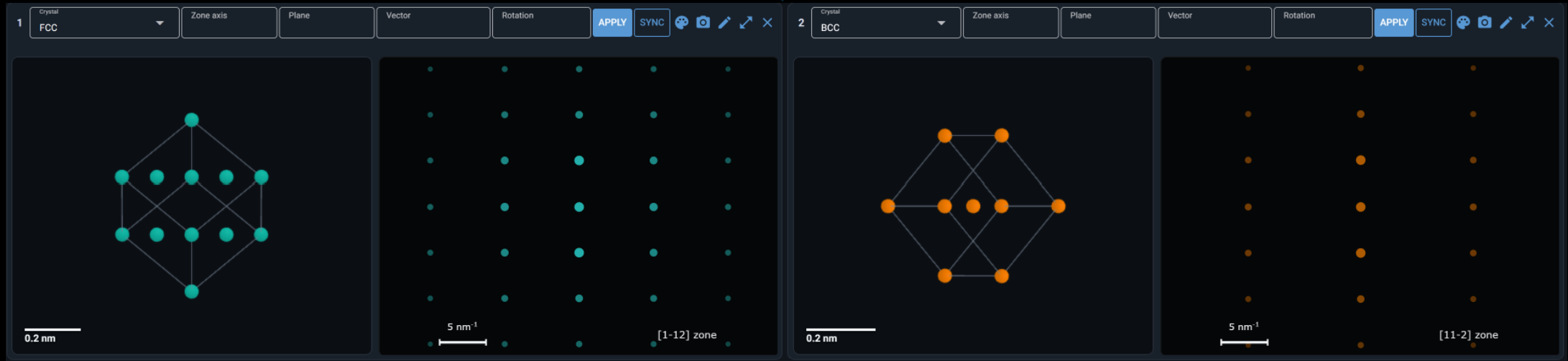
Features of the combo panel

- Bind crystal motion



Ex. if you go to FCC 1-12 zone in panel 1

BCC in panel 2 will do the same motion, and the resulting zone is 11-2



In this case, it means crystals 1 & 2 have the orientation relationship: $[110]_{\text{FCC}} // [111]_{\text{BCC}}$

Comparison of two simulation methods

Simulation method	Ewald Sphere Kinematic	Pymatgen TEMCalculator
Main purpose	More TEM-like kinematic spot visibility	Fast crystallographic zone-axis pattern
Uses voltage?	Yes, through electron wavelength and Ewald sphere radius	Only for wavelength calculation
Uses thickness?	Yes	No
Includes relrod / excitation error?	Yes	No
Includes Ewald sphere curvature?	Yes	No
Good for	Showing why spots appear/disappear with tilt/thickness	Clean ideal zone-axis indexing/intensity